Cryptography - Semester 2 Examinations 2017/2018

Question 1:

C Q R B R B J F N J T L R Y Q N A

2 16 17 1 17 1 9 5 13 9 19

encryption: $x \rightarrow x + K = y$

$E \rightarrow x + 9 = N$ so $K = 9$

Decryption: $x \rightarrow y - K \pmod{26}$

19 7 8 18 8 18 0 22 4 0 10

T H I S I S A W E A K

The first word = "THIS"

Question 2:

E S D C W N M H

4 18 3 2 22 13 12 7

$$A = \begin{bmatrix} 2 & 3 \\ 7 & 5 \end{bmatrix} \qquad A^{-1} = \frac{1}{15}\begin{bmatrix} 5 & -3 \\ -7 & 2 \end{bmatrix} = 7\begin{bmatrix} 5 & -3 \\ -7 & 2 \end{bmatrix} = \begin{bmatrix} 9 & 5 \\ 3 & 14 \end{bmatrix}$$

$\frac{1}{15} \equiv 7 \pmod{26}$

$$\begin{bmatrix} 9 & 5 \\ 3 & 14 \end{bmatrix}\begin{bmatrix} 4 \\ 18 \end{bmatrix} = \begin{bmatrix} 126 \\ 264 \end{bmatrix} \equiv \begin{bmatrix} 22 \\ 4 \end{bmatrix} = \begin{bmatrix} W \\ E \end{bmatrix}$$

$$\begin{bmatrix} 9 & 5 \\ 3 & 14 \end{bmatrix}\begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 11 \\ 11 \end{bmatrix} \equiv \begin{bmatrix} 11 \\ 11 \end{bmatrix} = \begin{bmatrix} L \\ L \end{bmatrix}$$

$$\begin{bmatrix} 9 & 5 \\ 3 & 14 \end{bmatrix}\begin{bmatrix} 22 \\ 13 \end{bmatrix} = \begin{bmatrix} 263 \\ 248 \end{bmatrix} = \begin{bmatrix} 3 \\ 14 \end{bmatrix} = \begin{bmatrix} D \\ O \end{bmatrix}$$

$$\begin{bmatrix} 9 & 5 \\ 3 & 14 \end{bmatrix}\begin{bmatrix} 12 \\ 7 \end{bmatrix} = \begin{bmatrix} 143 \\ 134 \end{bmatrix} = \begin{bmatrix} 13 \\ 4 \end{bmatrix} = \begin{bmatrix} N \\ E \end{bmatrix}$$

First two letters: WE

Question 3:

Describes Kerckhoffs Principle:

When examining the security of a cryptosystem we presume that an adversary knows exactly what system is being used but not the actual key being used. In this way, the security now depends on the mathematical properties of the cryptosystem, plaintext and keys, rather than our ability to keep the encryption / decryption algorithms secret.

(b) The difference between symmetric cryptography and public key cryptography

A symmetric cryptosystem consists of a set of $\mathcal{P}$ (plaintexts), $\mathcal{C}$ (ciphertexts) and $\mathcal{K}$ (keys) with functions $enc_k(x): \mathcal{P} \to \mathcal{C}$ and $dec_k(x): \mathcal{C} \to \mathcal{P}$ for all $k \in \mathcal{K}$. This is the same for public key cryptography, however, for all $k \in K$:

- There exists $e \in K$ (publick key) and $d \in K$ (private key) such that $dec_d(enc_e(x)) = x$ for all $x \in \mathcal{P}$

(c) One basic property of the Enigma cipher which makes it significantly more secure than the Vigenere cipher against a ciphertext only attack.

Question 4:

Determine, with explanation, the size of the enciphering key space for an affine cipher $f_E: (\mathbb{Z}_p)^d \to (\mathbb{Z}_p)^d$, $v \to Av + B$ over an alphabet of p letters with p a prime.

key (a,b) : $b,a \in \mathbb{Z}_n^{\times} \times \mathbb{Z}_n$

so $\#$ keys $= \varphi(n) \cdot n = \varphi(p) \cdot p = (p-1) \cdot p = p^2 - p$

Question 5:

A 4-bit linear feedback shift register with connection polynomial: $c(x) = 1 + x + x^3$ is used to produce a pseudo random sequence of binary digits $s_0 = 1$, $s_1 = 0$, $s_2 = 1$, $s_3 = 1$.

t	s_{t+2} (x)	s_{t+1} (x^2)	s_t (x^3)
0	1	0	1
1	1	1	0

t	s_{t+3} (x)	s_{t+2} (x^2)	s_{t+1} (x^3)	s_t (x^4)
0	1	1	0	1
1	1	1	1	0
2	0	1	1	1

2	1	1	1
3	0	1	1
4	0	0	1
5	1	0	0
6	0	1	0
7	1	0	1 ← initial state

So the least period of the sequence = 7.

3	1	1	1	1
4	0	1	1	1
5	1	0	1	1
6	0	1	0	1
7	0	0	1	0
8	1	0	0	1
9	1	1	0	0
10	1	1	1	0
11	0	1	1	1
12	1	0	1	1

Question 6:

(a)(i) Explain two-factor authentication:

Two-factor authentication (2FA) is a type of multi-factor authentication. It is a method of confirming users' claimed identities by using a combination of two different factors 1) something they know, 2) something they have or 3) something they are. A good example of this is withdrawing money from ATM, you need 1) bank card and 2) a pin.

(ii) Explain forward security:

Forward security is a feature of specific key agreement protocols that give assurances your session keys will not be compromised even if the private key of the server is compromised. By generating a unique session key for every session a user initiates, even the compromise of a single session key will not affect any data other than that exchanged in the specific session protected by that particular key.

(iii) Charmichael number.

A composite number $n \geq 4$ is a charmichael number iff $a^{n-1} \equiv 1 \pmod{n}$ for all $a \in \mathbb{Z}$ with $\gcd(a,n)=1$.

(b) Prove $|\mathbb{K}| \geq |\mathcal{C}| \geq |\mathcal{P}|$

- Pick $K \in \mathbb{K}$. Since $d_K(e_K(x)) = x$ for all $x \in \mathcal{P}$, e_K is injective from $\mathcal{P} \to \mathcal{C}$. Hence $|\mathcal{P}| \leq |\mathcal{C}|$.
- Let $x \in \mathcal{P}$ and $y \in \mathcal{C}$ then $\text{prob}(Y=y \mid X=x) = \text{Prob}(Y=y) > 0$

$\forall x \in \mathcal{P}, \forall y \in \mathcal{C} \ \exists K \in \mathcal{K}, y = e_K(x)$

Hence: given any $x \in \mathcal{P}$, the function $K \to \mathcal{C}$, $K \to e_K(x)$ is surjective

$|\mathbb{K}| \geq |\mathcal{C}|$ and so $|\mathbb{K}| \geq |\mathcal{C}| \geq |\mathcal{P}|$

Question 7:

(a)

0	0	1	0	0	0	1	1	0	1	0	1	0	1	1	1	p
1	0	1	0	0	1	1	1	0	1	0	0	1	1	1	0	k
1	0	0	0	0	1	0	0	0	0	0	1	1	0	0	1	e

$s_0 = 1, \quad s_1 = 0, \quad s_2 = 1$

length $= 3 \qquad s_L = 3 \qquad s_{2L-1} = 5$

$s_3 = c_1 s_2 + c_2 s_1 + c_3 s_0 = c_1(1) + c_2(0) + c_3(1) = 0 \Rightarrow c_1 + c_3 = 0$

$s_4 = c_1 s_3 + c_2 s_2 + c_3 s_1 = c_1(0) + c_2(1) + c_3(0) = 0 \Rightarrow c_2 = 0$

$s_5 = c_1 s_4 + c_2 s_3 + c_3 s_2 = c_1(0) + c_2(0) + c_3(1) = 1 \Rightarrow c_3 = 1$

$$e(x) = 1 + c_1 x + c_2 x^2 + c_3 x^3$$
$$e(x) = 1 + x + x^3$$

(b) Describe the A5/1 stream cipher which is used to encrypt the on-air traffic in the GSM Mobile phone networks.

A5/1 (GSM / 2G mobile phones) is based on 3 LSFR's of lengths 19, 22, and 23 respectively where key streams are added (in $\mathbb{Z}_2$). However, "clocking bits" are used to decide which registers are clocked at each step. The first attack was proposed in 1994. It was completely broken according to 2013 NSA leaks.

Question 8:

(a) State Euler's Totient Theorem:

Let $n \geq 1$ and let $a \in \mathbb{Z}$ with $\gcd(a,n) = 1$. Then $a^{\varphi(n)} = 1 \pmod n$

- Since $n = pq$, where p and q are two different primes $p \neq q$, the one congruence $a^{ed} = a \pmod n$ is equivalent to the two simultaneous congruences $a^{ed} = a \pmod p$ and $a^{ed} = a \pmod q$.

1. $p \mid a$: Then $a^{ed} = 0 \equiv a \pmod p$
2. $p \nmid a$: since $ed = 1 \pmod{(p-1)(q-1)}$ there exists an integer $x > 0$ with $ed = 1 + (p-1)(q-1)x$.

- By Fermat's little theorem: $a^{p-1} = 1 \pmod p$ and thus $a^{(p-1)(q-1)x} = (a^{p-1})^{(p-1)x} \equiv 1 \pmod p$

Hence $a^{ed} = a^{1+(p-1)(q-1)x}$

$= a \cdot a^{(p-1)(q-1)x} \equiv \cdot 1 = a \pmod p$

In the same way $a^{ed} \equiv 1 \pmod q$ so $(a^e)^d = 1 \pmod N$

(b) Describe Fermats primality test:

- Input: An integer $n \geq 2$
- Output: message "n is definitely composite" or "n is probably prime".
- Pick a random integer a from $1, \ldots, n-1$
- If a^{n-1} $1 \pmod n$, then n is definitely composite
- Otherwise, if n passes the above test for "several" random values of a, then report that n is probably prime.

Proposition: Proof:

Define a function $f: \mathbb{Z}_n^\times \to \mathbb{Z}_n^\times$, $x \to ax \bmod n$

since $\gcd(a,n) = 1$, f is well-defined and bijective

let $X = \{x \in \mathbb{Z}_n^\times : x^{n-1} \equiv 1 \pmod n\}$ and $Y = \{x \in \mathbb{Z}_n^\times : x^{n-1} \not\equiv 1\}$

claim $f(x) \subseteq Y$. Indeed, if $x \in X$ then $f(x)^{n-1} = (ax)^{n-1} \equiv a^{n-1}x^{n-1}$

Thus since f is injective $|X| = |f(x)| \leq Y$.

The claim follows since $|\mathbb{Z}_n^\times| = |X| + |Y|$

Question 9:

(a) Diffie-Hellman key exchange

$\mathbb{Z}_{71}^\times$ and $g = 7$

1 • Alices chooses $a = 6$ and computes $g^a = 117649 \equiv 2 \bmod 71$

2 • Bob chooses $b = 13$ and computes $g^b = 28 \bmod 71$

4 • Alice recieves $B = 0$ and computes $B^a = 28^6 \equiv 27 \bmod 71$

8 • Bob recieves $A = 2$ and computes $A^b \equiv 2^{13} \equiv 27 \bmod 71$

$13 = 1 + 4 + 8 = 2^0 + 2^2 + 2^3$

$7^{13} = (7^1 \cdot 7^4) 7^8$ 8192

$7^1 = 7, \quad 7^4 = 58, \quad 7^8 = (7^4)^2 = 58^2 = 3364 = 27$

$7^{13} = (7 \cdot 58)(27) = (51)(27) = 1377 = 28$ 481890304

So 27 is their shared key. 6787187

(b) Describe the operations of addition and subtraction which give E the structure of an abelian group.

- If $P = O$ then we define $-P = O$ and $P + Q = Q$

If $Q = O$, we define $P + Q = P$ (O = identity element).

Henceforth, suppose that $P \neq O \neq Q$.

(learn the rest) (in copy).

Question 10:

(b) Shannon's Theorem:

Let $|\mathcal{K}| = |\mathcal{C}| = |\mathcal{P}|$, then our cryptosystem is perfectly secure if the following holds:

- $\text{Prob}(K \in \mathcal{K}) = \frac{1}{|\mathcal{K}|}$ for all $K \in \mathcal{K}$.
- $\forall x \in \mathcal{P}, \forall y \in \mathcal{C} \; \exists K \in \mathcal{K} \;\; e_K(x) = y$.

One time pads are an example of perfectly secure cryptosystems. This is when $\mathcal{P} = \mathcal{C} = \mathcal{K} = \mathbb{Z}_n^d$ with $e_K(x) = x + k$ and $d_K(y) = y - k$. and uniformly chosen keys. By Shannons theorem this is perfectly secure. Condition (a) is satisfied by construction, and for $x, y \in \mathbb{Z}_n^d$, the unique $K \in \mathbb{Z}_n^d$ with $x + K = y$ is $k = y - x$.